

Homework Set 3, CPSC 6430/4430

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Due 03/31/2026, 11:59PM EST

PCA

Please refer to Jupyter Notebook.

Naive Bayes

The table below provides a training dataset containing sixteen observations. Now please determine by Naive Bayes whether the new coming data is Male (M) or Female (F) given ‘Basketball, 6 feet (height), 150lbs, 4 inches hair, 10 inches footsize and Drinking’.

Sports	Height (feet)	Weight (lbs)	Hair (inches)	Footsize (inches)	Drinking	Class
Basketball	6	145	6	11	No	F
Tennis	5.6	150	8	11	Yes	M
Basketball	6.4	170	3	13	Yes	M
Basketball	6.2	150	7	12	No	M
Basketball	6	140	6	11	Yes	F
Tennis	6.4	130	7	12	No	F
Soccer	5.8	140	4	12	Yes	M
Soccer	6.2	130	7	12	No	F
Soccer	6	150	4	10	Yes	M

Naive Bayes Solution

We want to classify the new data point

$$X = (\text{Basketball}, 6, 150, 4, 10, \text{Yes})$$

as either Male (M) or Female (F).

Step 1: Class Priors

From the table, there are 9 observations total.

$$P(M) = \frac{5}{9}, \quad P(F) = \frac{4}{9}$$

since there are 5 males and 4 females.

Step 2: Conditional Probabilities

Using Naive Bayes, we compute:

$$P(M | X) \propto P(M) P(\text{Basketball} | M) P(6 | M) P(150 | M) P(4 | M) P(10 | M) P(\text{Yes} | M)$$

$$P(F | X) \propto P(F) P(\text{Basketball} | F) P(6 | F) P(150 | F) P(4 | F) P(10 | F) P(\text{Yes} | F)$$

For class M

Among the 5 male observations:

$$P(\text{Basketball} \mid M) = \frac{2}{5}$$

$$P(6 \mid M) = \frac{1}{5}$$

$$P(150 \mid M) = \frac{3}{5}$$

$$P(4 \mid M) = \frac{2}{5}$$

$$P(10 \mid M) = \frac{1}{5}$$

$$P(\text{Yes} \mid M) = \frac{4}{5}$$

Thus,

$$P(M \mid X) \propto \frac{5}{9} \cdot \frac{2}{5} \cdot \frac{1}{5} \cdot \frac{3}{5} \cdot \frac{2}{5} \cdot \frac{1}{5} \cdot \frac{4}{5}$$

$$P(M \mid X) \propto \frac{16}{9375} \approx 0.0017067$$

For class F

Among the 4 female observations:

$$P(\text{Basketball} \mid F) = \frac{2}{4} = \frac{1}{2}$$

$$P(6 \mid F) = \frac{2}{4} = \frac{1}{2}$$

$$P(150 \mid F) = \frac{0}{4} = 0$$

$$P(4 \mid F) = \frac{0}{4} = 0$$

$$P(10 \mid F) = \frac{0}{4} = 0$$

$$P(\text{Yes} \mid F) = \frac{1}{4}$$

So,

$$P(F \mid X) \propto \frac{4}{9} \cdot \frac{1}{2} \cdot \frac{1}{2} \cdot 0 \cdot 0 \cdot 0 \cdot \frac{1}{4} = 0$$

Step 3: Classification

Since

$$P(M | X) > P(F | X),$$

the new data point is classified as

Male (M)

Final Answer

The new observation is classified as Male (M).
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